Review of Stochastic Optimal Control in Continuous Time

Parts of this lecture are based on notes from Chris Edmond (Melbourne) and Ben Moll (Princeton)

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Deterministic control

- Now consider the control problem

$$v(x(t),t) = \max_{u(\cdot)} \int_t^\infty e^{-\rho(s-t)}\, \pi(x(s),u(s))\, ds, \qquad t \geq 0$$

with state variable x, control u, subject to the law of motion

$$\dot{x}(t) = g(x(t),u(t)), \qquad x(0) = x_0 \text{ given}$$

- Value function satisfies the continuous time Bellman equation (also called Hamilton-Jacobi-Bellman equation).

$$\rho v(x,t) = \max_u \left[\pi(x,u) + \frac{\partial v}{\partial x} g(x,u)\right] + \frac{\partial v}{\partial t}$$

where the maximand is the Hamiltonian function

$$H(x,u,\lambda) = \pi(x,u) + \lambda g(x,u)$$

with costate variable $\lambda = \partial v/\partial x$

Heuristic derivation of the Bellman equation

- Discrete time analogue with period length $\Delta t > 0$, $t' = t + \Delta t$, $x' = x + \Delta x$ satisfies

$$v(x,t) = \max_u \left[\pi(x,u)\,\Delta t + \frac{1}{1+\rho\Delta t} v(x',t') \right]$$

- Multiplying both sides by $(1+\rho\Delta t)$ and subtracting $v(x,t)$

$$\rho v(x,t)\Delta t = \max_u \left[\pi(x,u)\,(1+\rho\Delta t)\Delta t + v(x',t') - v(x,t) \right]$$

- Write out the change in the value function as follows (similar to a first order Taylor Expansion)

$$v(x',t') - v(x,t) = v(x',t') - v(x,t') + v(x,t') - v(x,t)$$

$$= \frac{v(x',t') - v(x,t')}{\Delta x}\Delta x + \frac{v(x,t') - v(x,t)}{\Delta t}\Delta t$$

Heuristic derivation of the Bellman equation

- Now use the state transition equation $\Delta x = g(x,u)\Delta t$ to write

$$v(x',t') - v(x,t) = \frac{v(x',t') - v(x,t')}{\Delta x} g(x,u)\Delta t + \frac{v(x,t') - v(x,t)}{\Delta t}\Delta t$$

- Plug change in value function back into Bellman equation

$$\rho v(x,t)\Delta t = \max_u \left[\pi(x,u)\,(1+\rho\Delta t)\Delta t + \frac{v(x',t') - v(x,t')}{\Delta x} g(x,u)\Delta t\right] + \frac{v(x,t') - v(x,t)}{\Delta t}\Delta t$$

- Divide both sides by $\Delta t > 0$ and take limit as $\Delta t \to 0$ to get

$$\rho v(x,t) = \max_u \left[\pi(x,u) + \frac{\partial v}{\partial x} g(x,u)\right] + \frac{\partial v}{\partial t}$$

Stochastic Bellman Equations

- Consider the value function

$$v(x(t),t) = \max_u \mathbb{E}\Big[\int_t^\infty e^{-\rho(s-t)} \pi(x(s),u(s))\, ds \,\Big|\, x(t)\Big], \qquad t \geq 0$$

  where $x(t)$ follows a stochastic process

- What is the Bellman equation for this problem?

- Discrete time analogue with period length $\Delta t > 0$, $t' = t + \Delta t$, $x' = x + \Delta x$ satisfies

$$v(x,t) = \max_u \pi(x,u)\Delta t + \frac{1}{1+\rho\Delta t}\mathbb{E}[v(x',t') \,|\, x]$$

Stochastic Bellman Equation

- Multiplying both sides by $(1+\rho\Delta t)$ and subtracting $v(x,t)$ gives

$$\rho v(x,t)\Delta t = \max_u \pi(x,u)\Delta t(1+\rho\Delta t) + \mathbb{E}[v(x',t') - v(x,t)\,|\,x]$$

- Again write out the change in the value function

$$v(x',t') - v(x,t) = v(x',t') - v(x,t') + v(x,t') - v(x,t)$$

$$= v(x',t') - v(x,t') + \frac{v(x,t') - v(x,t)}{\Delta t}\Delta t$$

- The latter change is *deterministic* and can be pulled outside of the expectation, so

$$\rho v(x,t)\Delta t = max_u\{\pi(x,u)\Delta t(1+\rho\Delta t) + \mathbb{E}[v(x',t') - v(x,t')\,|\,x] + \frac{v(x,t') - v(x,t)}{\Delta t}\Delta t\}$$

Stochastic Bellman Equation

The treatment $\mathbb{E}[v(x', t') - v(x, t') \mid x]$ depends on specific assumptions of the stochastic process of x.

We investigate continuous time optimal control under two cases

- Poisson Process: Turns out to be very convenient in continuous time because of its step-by-step nature
- Diffusion Process: Relies on Ito's Lemma, elegant math

Poisson process

- Continuous time stochastic process $x(t)$ with $x(0) = 0$ and

  (i) stationary independent increments
  (ii) increments have Poisson distribution with intensity $\lambda > 0$

- Increments $x(t') - x(t)$ are independent r.v.'s, for all t, t'

- Increments $\Delta x = x(t + \Delta t) - x(t)$ have Poisson distribution

$$\text{Prob}[\Delta x = k] = \frac{(\lambda \Delta t)^k e^{-\lambda \Delta t}}{k!}$$

  (depends only on period length $\Delta t > 0$, independent of actual t)

- Sample paths are discontinuous, a 'counting process'

Properties of Poisson process

- Write the distribution of the increment from date s to $s+t$

$$\text{Prob}[\,\{x(t+s)-x(s)\}\,=k]=\frac{(\lambda t)^k e^{-\lambda t}}{k!}$$

  that is, a Poisson distribution with parameter λt

- In particular, taking $s=0$ (and using $x(0)=0$) we simply have

$$\text{Prob}[x(t)=k]=\frac{(\lambda t)^k e^{-\lambda t}}{k!}$$

- So, using standard properties of the Poisson distribution, the process $x(t)$ has moments

$$\mathbb{E}[x]=\text{Var}[x]=\lambda t$$

Properties of Poisson process

- Probability that $\Delta x = 1$ over period of length Δt is

$$\text{Prob}[\Delta x = 1] = (\lambda \Delta t) e^{-\lambda \Delta t}$$

- Probability process does not change, $\Delta x = 0$, over same period

$$\text{Prob}[\Delta x = 0] = e^{-\lambda \Delta t}$$

(note link to exponential distribution...)

- These can be written

$$\text{Prob}[\Delta x = 1] = \lambda \Delta t + o(\Delta t)$$
$$\text{Prob}[\Delta x = 0] = 1 - \lambda \Delta t + o(\Delta t)$$

where $o(\Delta t)/\Delta t \to 0$ as $\Delta t \to 0$ and moreover where

$$\text{Prob}[\Delta x > 1] = o(\Delta t)$$

Bellman equation when state is Poisson

- Now write out the expectation

$$\mathbb{E}[v(x',t') - v(x,t') \,|\, x] = \sum_{\Delta x=0}^{\infty} [v(x+\Delta x,t') - v(x,t')]\text{Prob}[\Delta x]$$

$$\begin{aligned} = \quad & [v(x+0,t') - v(x,t')]\text{Prob}[\Delta x = 0] \\ + \; & [v(x+1,t') - v(x,t')]\text{Prob}[\Delta x = 1] \\ + \; & [v(x+2,t') - v(x,t')]\text{Prob}[\Delta x = 2] \\ + \; & \ldots \end{aligned}$$

- Hence for the Poisson process

$$\mathbb{E}[v(x',t') - v(x,t') \,|\, x] = [v(x+1,t+\Delta t) - v(x,t+\Delta t)]\lambda\Delta t + o(\Delta t)$$

Bellman equation when state is Poisson

- Plug this back into the Bellman equation

$$\begin{aligned}\rho v(x,t)\Delta t = &\pi(x)\Delta t(1+\rho\Delta t)\\ &+[v(x+1,t+\Delta t)-v(x,t+\Delta t)]\lambda\Delta t+o(\Delta t)\\ &+\frac{v(x,t')-v(x,t)}{\Delta t}\Delta t\end{aligned}$$

- Divide both sides by $\Delta t > 0$ and take limit as $\Delta t \to 0$ to get

$$\rho v(x,t) = \pi(x) + \lambda(v(x+1,t)-v(x,t)) + \frac{\partial v}{\partial t}$$

A more general Birth and Death process

- Generalize Poisson to a few competing events
- The probability distribution of the waiting time until the next "birth" is Exponential λ
- The probability distribution of the waiting time until the next "death" is Exponential μ
- The arrival rate of the next event is $\lambda + \mu$. Conditional on the event, the prob of "birth" is $\frac{\lambda}{\lambda+\mu}$, and the prob of "death" is $\frac{\mu}{\lambda+\mu}$.
- Thus we have the B-D process

$$\begin{aligned} \text{Prob}[\Delta x = 1] &= \lambda \Delta t + o(\Delta t) \\ \text{Prob}[\Delta x = -1] &= \mu \Delta t + o(\Delta t) \\ \text{Prob}[\Delta x = 0] &= 1 - \lambda \Delta t - \mu \Delta t + o(\Delta t) \end{aligned}$$

where $o(\Delta t)/\Delta t \to 0$ as $\Delta t \to 0$ and $\text{Prob}[|\Delta x| > 1] = o(\Delta t)$

Bellman equation when state is B-D

- Plug this back into the Bellman equation

$$\begin{aligned}\rho v(x,t)\Delta t = &\pi(x)\Delta t(1+\rho\Delta t)\\ &+[v(x+1,t+\Delta t)-v(x,t+\Delta t)]\lambda\Delta t+o(\Delta t)\\ &+[v(x-1,t+\Delta t)-v(x,t+\Delta t)]\mu\Delta t+o(\Delta t)\\ &+\frac{v(x,t')-v(x,t)}{\Delta t}\Delta t\end{aligned}$$

- Divide both sides by $\Delta t > 0$ and take limit as $\Delta t \to 0$ to get

$$\rho v(x,t) = \pi(x)+\lambda(v(x+1,t)-v(x,t))+\mu(v(x-1,t)-v(x,t))+\frac{\partial v}{\partial t}$$

Continuous time Markov Chains

- Both Poisson process and B-D process are special cases of a Continuous time Markov Chain

- In CTMC, time is perturbed by Exponentially distributed *holding times* in each state i

- The transition probability is defined by

$$P_{ij}(t) = P(X(t+s) = j | X(s) = i)$$

  which is a function of t (recall the previous examples).

- As a result, unlike discrete time case, it is not enough to specify a one-step transition matrix

- However, the transition probability matrices should be chosen to satisfy Chapman-Kolmogorov equation

$$P_{ij}(t+s) = \sum_k P_{ik}(t) P_{kj}(s)$$

Kolmogorov Equations

We now derive a set of differential equations, called Kolmogorov Equations, to characterize $p_{ij}(t)$. We start with the "Backward Equations". Let $h = \Delta t$ be a small positive amount of time.

- We start with evaluating at $t + h$, condition on $X(h)$

$$\begin{aligned} p_{ij}(t+h) &= P(X(t+h) = j | X(0) = i) \\ &= \sum_k P(X(t+h) = j | X(h) = k) P(X(h) = k | X(0) = i) \\ &= \sum_k P(X(t) = j | X(0) = k) P(X(h) = k | X(0) = i) \\ &= \sum_k p_{kj}(t) P(X(h) = k | X(0) = i) \end{aligned}$$

  where the third equality follows time-homogeneity

- Define transition rate from state i to j as exponential q_{ij} and its holding time at i is exponential $v_i = \sum_j q_{ij}$. Upon leaving state i, its probability of going j is q_{ij}/v_i.

Kolmogorov Equations

- Back to our earlier derivation, we can now write $p_{ij}(t+h)$ separating out $k \neq i$

$$p_{ij}(t+h) = p_{ij}(t)(1 - v_i h + o(h)) + \sum_{k \neq i} p_{kj}(t)(q_{ik}h + o(h))$$

$$\frac{p_{ij}(t+h) - p_{ij}(t)}{h} = -v_i p_{ij}(t) + \sum_{k \neq i} p_{kj}(t)(q_{ik} + \frac{o(h)}{h})$$

- Let $h \to 0$, the LHS approaches $p_{ij}^{'}(t)$, we have

$$p_{ij}^{'}(t) = \sum_{k \neq i} q_{ik} p_{kj}(t) - v_i p_{ij}(t)$$

- This is called Kolmogorov's Backward Equations. If instead, we condition on $X(t)$ (in the third equality above), we have the Kolmogorov Forward Equations

$$p_{ij}^{'}(t) = \sum_{k \neq j} q_{kj} p_{ik}(t) - v_j p_{ij}(t)$$

Kolmogorov Equations: Examples

We typically use the Kolmogorov Forward Equations to characterize B-D process

- Discrete states $i = 0, 1, 2, 3, ..., n, ...$
- We define $q_{kj} = \lambda, j = i + 1$, $q_{kj} = \mu, j = i - 1$, $q_{kj} = 0, |j - k| > 1$
- It is easy to show that the KFE is

$$\dot{p}_n(t) = \lambda p_{n-1}(t) + \mu p_{n+1}(t) - (\lambda + \mu) p_n(t)$$
$$\dot{p}_0(t) = \mu p_1(t) - \lambda p_0(t)$$

- The second equation indicates 0 is the lowest state possible.

Infinitesimal Generator Matrix

- Also called transition rate matrix, an IGM describes the rate a continuous markov chain moves between states
- It is the counterpart of one-step transition matrix for discrete-time MC
- The system of KFEs can be written in matrix form (of states)

$$\dot{\mathbf{P}}(t) = \mathbf{P}(t)\mathbf{G}$$

- $\mathbf{G}$ has $-v_j$ (or $-v_i$) on the diagonal, and q_{kj} off the diagonal.
- Finally, we have the boundary condition $\mathbf{P}(0) = \mathbf{I}$.
- The nice feature of the IGM is that both KFE and KBE imply that

$$\mathbf{P}(t) = \sum_{n=0}^{\infty} \frac{(t\mathbf{G})^n}{n!} \equiv e^{t\mathbf{G}}$$

where the right-hand side is matrix exponential (can be evaluated in MATLAB easily).

Kolmogorov Equations: Closed-form Solution

- We treat these cases as a "Cook Book"

- Define probability generating function for a discrete RV X_t as $F_X(t,z) = E[z^{X_t}] = \sum_{n=0}^{\infty} P[X_t = n] z^k$

- With initial condition n_0, the following cases' PGF can be analytically solved
  - "Birth process" with rate $n\lambda$: $F_X(z,t) = [\frac{z}{z+(1-z)e^{-\lambda t}}]^{n_0}$.
  - "B-D process" with rate $n\lambda$ and $n\mu$: $F_X(z,t) = [\frac{(\mu-\lambda z)e^{-t(\lambda-\mu)}-\mu(1-z)}{(\mu-\lambda z)e^{-t(\lambda-\mu)}-\lambda(1-z)}]^{n_0}$
  - "B-D process" with rate λ and $n\mu$: $F_X(z,t) = e^{-\lambda(1-z)(1-e^{-\mu t}/\mu}[1-(1-z)e^{-\mu t}]^{n_0}$

- Taylor series expansion of $F_X(z,t)$ around $z=0$ yields $p_n(t;n_0)$ as the coefficients

$$p_n(t;n_0) = \frac{1}{n!}\frac{\partial^n H(z,t;n_0)}{\partial z^n}|_{z=0}$$

and $p_0(t;n_0) = H(0,t;n_0)$

Standard Brownian motion

- Continuous time stochastic process $w(t)$ with

  (i) continuous sample paths (no jumps)
  (ii) stationary independent increments
  (iii) distribution $w(t) \sim N(0, t)$

- That is, over period of length Δt the change has distribution

$$\Delta w = \varepsilon_t \sqrt{\Delta t}, \qquad \varepsilon_t \sim \text{IID } N(0,1)$$

- In limit as period length becomes infinitesimally small, write $dw = \varepsilon_t \sqrt{dt}$ so that

$$\mathbb{E}[dw] = \mathbb{E}[\varepsilon_t \sqrt{dt}] = 0$$

$$\text{Var}[dw] = \mathbb{E}[\varepsilon_t^2 dt] = dt$$

- Sometimes referred to as "continuous time random walk" (zero mean, but blows up overtime).

Generalize Brownian Motion

- A more general Diffusion Process is

$$dX_t = \mu(t, X_t)dt + \sigma(t, X_t)dW_t$$

  where W_t follows Brownian Motion.

- We call $\mu(t, X_t)$ drift, $\sigma(t, X_t)$ diffusion. These two terms give a lot of flexibility to construct stochastic process in continuous time.

  - Ornstein-Uhlenbeck Process: $\mu(t, X_t) = \rho(\mu - X_t)$, $\sigma(t, X_t) = \sigma$. This is a counterpart to $AR(1)$ in discrete time. It is stationary and mean-reverting towards μ.

  - Square-Root Process: $\mu(t, X_t) = \rho(\mu - X_t)$, $\sigma(t, X_t) = \sigma\sqrt{X_t}$. Like O-U, it is mean-reverting to μ, but with volatility proportional to $\sqrt{X_t}$.

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Ito's Lemma

- Think of some state variable in HJB is a diffusion process. If value function is well-behaved, can we say something about $E[\frac{dV}{dt}]$?

- Ito's Lemma provides powerful results. Let $dX_t = \mu(t, X_t)dt + \sigma(t, X_t)dW_t$ be a diffusion process and if $f(X_t)$ is any twice-differentiable function, then

$$df(X_t) = [\mu(t, X_t)f'(X_t) + \frac{1}{2}\sigma(t, X_t)^2 f''(X_t)]dt + \sigma(t, X_t)f'(X_t)dW_t$$

- Multivariate case can be defined similarly. Importantly, it says $f(X_t)$ itself is also a diffusion process.

- One quick application: Geometric Brownian Motion $Y_t = e^{X_t}$ where X_t is a Brownian Motion with constant drift μ and diffusion σ, i.e. $dX_t = \mu dt + \sigma dW_t$. Using Ito's Lemma

$$dY_t = [\mu e^{X_t} + \frac{1}{2}\sigma^2 e^{X_t}]dt + \sigma e^{X_t} dW_t \equiv [\mu + \frac{1}{2}\sigma^2]Y_t dt + \sigma Y_t dW_t$$

  Hence $dlnY_t \equiv dY_t/Y_t = [\mu + \frac{1}{2}\sigma^2]dt + \sigma dW_t$.

Bellman equation when state is Diffusion Process

- Recall that the stochastic

$$\rho v(x,t)\Delta t = max_u\, \pi(x,u)\Delta t(1+\rho\Delta t) + \mathbb{E}[v(x',t') - v(x,t') \,|\, x] + \frac{v(x,t') - v(x,t)}{\Delta t}\Delta t$$

- Divide both side by Δt and let $\Delta t \to 0$, we have

$$\rho v(x,t) = max_u\, \pi(x,u) + \frac{\mathbb{E}[dv(x,t)]}{dt}$$

given that $\frac{\partial v(x,t)}{\partial t} = 0$.

- Apply Ito's Lemma, we have

$$\frac{\mathbb{E}[dv(x,t)]}{dt} = \mu(x,u,t)\frac{\partial v(x,t)}{\partial x} + \frac{1}{2}\sigma(x,u,t)^2\frac{\partial^2 v(x,t)}{\partial x^2}$$

Multivariate Case

- For convenience, we now take out t.

- The diffusion process is general enough, so we can easily incorporate both endogenous and exogenous states

$$dx_1 = g(x_1, x_2, u)dt$$
$$dx_2 = \mu(x_2)dt + \sigma(x_2)dW$$

- Then we have

$$\frac{\mathbb{E}[dv(x)]}{dt} = g(x_1, x_2, u)\frac{\partial v(x_1, x_2)}{\partial x_1} + \mu(x_2)\frac{\partial v(x_1, x_2)}{\partial x_2} + \frac{1}{2}\sigma(x_2)^2\frac{\partial^2 v(x_1, x_2)}{\partial x_2^2}$$

- Substitute in the value function

$$\rho v(x,t) = max_u\, \pi(x,u) + g(x_1, x_2, u)\frac{\partial v(x_1, x_2)}{\partial x_1} + \mu(x_2)\frac{\partial v(x_1, x_2)}{\partial x_2} + \frac{1}{2}\sigma(x_2)^2\frac{\partial^2 v(x_1, x_2)}{\partial x_2^2}$$

Kolmogorov Equations

- Similar to the case of continuous time MC, powerful results exist to characterize the evolution of the density of X, denoted as $p(x,t)$, if X itself is a diffusion process.

- If $dX_t = \mu(X_t,t)dt + \sigma(X_t,t)dW_t$, then

$$\frac{\partial p(x,t)}{\partial t} = -\frac{\partial}{\partial x}[\mu(x,t)p(x,t)] + \frac{1}{2}\frac{\partial^2}{\partial x^2}[\sigma(x,t)^2 p(x,t)]$$

- This KFE is also called "Fokker-Planck" equation.

- Again for the case with both endogenous and exogenous states, we have

$$\frac{\partial p(x_1,x_2,t)}{\partial t} = -\frac{\partial}{\partial x_1}[g(x_1,x_2,u)p(x,t)] - \frac{\partial}{\partial x_2}[\mu(x_2)p(x,t)] + \frac{1}{2}\frac{\partial^2}{\partial x_2^2}[\sigma(x_2)^2 p(x,t)]$$

Kolmogorov Equations: Example

- We now revisit Gabaix's argument of random growth
- Recall the proportional growth $S_{t+\Delta t} = \gamma_{t+\Delta t} S_t$. Assume that

$$\gamma_{t+\Delta t} = 1 + \mu \Delta t + \sigma \epsilon \sqrt{\Delta t}$$

  where $\epsilon \sim N(0, 1)$.
- Thus we have $S_{t+\Delta t} - S_t = (\mu \Delta t + \sigma \epsilon \sqrt{\Delta t}) S_t$, as $\Delta t \to 0$

$$dS_t = \mu S_t dt + \sigma S_t dW_t$$

  which is a geometric Brownian Motion.
- Similar to Gibrat's original formulation, this implies exploding distribution.

Kolmogorov Equations: Example

- A clever and natural "friction" to put in is to assume that if the process goes below S_m, then it is brought back to S_m (reflecting barrier).

- Now apply the Kolmogorov Forward Equation to the implied stationary distribution

$$0 = -\frac{\partial}{\partial x}[\mu x f(x)] + \frac{1}{2}\frac{\partial^2}{\partial x^2}[\sigma^2 x^2 f(x)]$$

- Can guess and verify that $f(x) = Cx^{-\zeta-1}$, where $\zeta = 1 - \mu/\sigma^2$. For mean to exist, need $\zeta > 1$ (i.e., $\mu < 0$).

- Pin down C with $1 = \int_{S_m} f(x)dx$, thus $C = S_m^{\zeta}$.

- Define mean size $\bar{S} = \int_{S_m} xf(x)dx = \frac{\zeta}{\zeta-1}S_m$. When $S_m in/\bar{S} \to 0$, then $\zeta \to 1$. The Zipf's Law obtains as friction becomes small.